

Daniel Díaz Reyes

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Education

University of Colorado Boulder | Boulder, CO

Expected Graduation Date: May 2027

Bachelor of Science in Electrical and Computer Engineering

GPA: 3.25

Technical Skills

- **Programming:** C/C++, Python, MATLAB, Linux, System Verilog
- **Embedded/Robotics:** ROS2, Arduino C, Raspberry Pi, RISC-V, STM32 (Cube IDE), IssacSim
- **Design:** Altium, KiCAD, SolidWorks, SIMetrix
- **Electrical:** Proficient in **soldering**, measurements using **oscilloscope** and **digital multimeter**, **PCB testing**
- **Language:** Fluent in **Spanish** and **English**; experienced in document translation

Relevant Coursework

Linear Algebra (Spring '25), Introduction to Probability (Spring '25), PCB Design and Manufacture (Spring '26), Linear Systems (Spring '26), Embedded Software (Spring '26), RTOS (Spring '26), Computer Organization (Spring '26)

Relevant Work Experience

Colorado School of Mines

Intelligent Robotics and Systems Laboratory | Golden, CO

March 2024 – August 2025

Undergraduate Research Fellow

- **Reverse-engineered** an industrial robot **communication stack** using network packet sniffing to analyze **system control** interfaces
- Analyzed **PCB layout**, read data sheets, and identified signal lines to reverse-engineer camera data interfaces; identified **Ethernet-based** camera data communication
- Developed a **prototype** for a camera video capture strategy with **Raspberry Pi 5** and **CSI camera** modules
- Developed **technical documentation** and **system flow diagrams** to support cross-lab collaboration and inform research on **autonomous systems** for snow-impacted communities

Colorado Fuel Cell Center | Golden, CO

April 2024 – February 2025

Research Intern

- Designed and implemented upgrades to a pressurized test stand using **SolidWorks**, enhancing **hydrogen membrane reactor** characterization for **industrial carbon reduction** in partnership with a local developer
- Interfaced with industrial developer, **Utility Global**, bi-weekly to review objectives and project progress
- Installed and commissioned **hardware modifications** at ~850 °C and ~6 atm, achieving full operational readiness and validating test stand performance
- **Calibrated** and **sized** six mass flow controllers to achieve airtight test stand operation and meet target residence times, optimizing performance and efficiency at target conditions
- **Finalist in the 2024 STEM Research Competition - Oral Presentation at the 2024 SHPE National Convention**

Engineering Projects Experience

Mines Battery Workforce Challenge Team

December 2023 - May 2025

- Inter-collegiate **Stellantis-sponsored** competition to design a full-sized battery pack for a **RAM ProMaster EV**
- Post-processed and analyzed battery performance data, using **Excel** and **MATLAB**
- Designed **CAD** models for the battery disconnect unit (**BDU**) in **SolidWorks**, supporting subsystem integration
- Developed **PCBs** to optimize connections between **BDU** components, successfully providing power distribution

Arduino Drone Prototype

November 2024 - December 2024

- Built Arduino-based drone with accelerometer stabilization and custom transmitter code using **Arduino C**
- Designed and **3-D printed** a lightweight **PLA+** chassis using **SolidWorks**, weighing a total of ~240g

Leadership and Outreach

Society of Hispanic Professional Engineers (SHPE)

September 2023 - Present

- Served as Outreach Co-Coordinator for the Colorado School of Mines Chapter, promoting STEM education
- Led **Medtronic** partnership for Northglenn's "**Noche de Ciencias**" event to boost student STEM exposure
- Co-planned the Mines SHPE chapter's "**Día de Ciencias Event**" with the goal to foster community engagement
- Taught high school students to obtain college equivalency credit for "Cornerstone Design I"